Customer Churn Prediction Using Machine Learning

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1. Introduction

Customer churn refers to customers who stop using a company's services. This project predicts telecom customer churn using machine learning to help improve customer retention.

2. Objectives

- Predict customer churn.
- Analyze customer behavior.
- Perform data cleaning and preprocessing.
- Build and evaluate a Logistic Regression model.

3. Dataset

Telco Customer Churn Dataset containing customer demographics, services, contract details, payment methods, monthly charges, total charges, and churn status.

4. Tools

Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn.

5. Data Preprocessing

Handled missing values, converted TotalCharges to numeric, removed customerID, encoded categorical variables, and prepared the data for machine learning.

6. Exploratory Data Analysis

Created visualizations for churn distribution, contract type, payment method, monthly charges, and correlation heatmap.

7. Machine Learning Model

Algorithm: Logistic Regression
Training/Test Split: 80/20
Accuracy: **78.75%**

8. Key Findings

- Month-to-month contracts have the highest churn.
- Electronic check users are more likely to churn.
- Higher monthly charges increase churn probability.
- Long-term contracts improve retention.

9. Business Recommendations

Promote long-term contracts, provide loyalty rewards, improve customer support, and offer targeted retention plans.

10. Conclusion

The project successfully demonstrated an end-to-end data analysis and machine learning workflow for predicting telecom customer churn with 78.75% accuracy.